

“It will take India at least five years to realise the full impact of smart grid”

Satish Aggarwal, member, IEEE-SA Board of Governors, and senior programme manager, United States Nuclear Regulatory Commission, shares with Uma Bansal, executive editor at EFY, the major issues in smart grid operation and implementation and how IEEE-SA is positioned to lead the smart grid initiative in India



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What are the goals of India's smart grid vision?

India's smart grid efforts primarily concern three main issues: Increased load needs which cannot be met by the present supply and hence result in frequent brownouts, the drive to electrify a large segment of its rural population, and the need to optimise electrical usage by being able to manage loads and mitigate operating inefficiencies (the losses in the system, both financial and technical, are amongst the highest in the world).

India's Smart Grid vision as expressed by the India Smart Grid Forum has five fundamental objectives: End of load sharing (with peak load shifting through a combination of direct control and differential pricing), reliable power (through robust systems with self-healing capabilities), cheaper power (through dramatic reduction in aggregate technical and commercial (AT&C) losses and real-time monitoring of load sources), shifting the peak away from costly power (through better utilisation of assets) and more sustainable power (through integration of green and renewable resources at a massive scale).

Greater use of information and communication technologies needs to be encouraged to make three fundamental improvements to India's existing power grid: Advanced metering to reduce AT&C losses; automation to measure and control the flow of power to/from consumers on a near-real-time basis and improve the system reliability; and moving to a smart grid to intelligently manage loads, congestion and shortfall.

Do you see these achievable? What are major issues in smart grid operation and implementation, particularly for India?

India is on the right path. The government of India's Restructured Accelerated Power Development and Reform Programme (R-APDRP) is a good step for successful implementation of the smart grid in the country. The programme is designed to take three to five years to implement across parts of India.

However, successful and complete implementation of the smart grid is not going to be an easy task as the Indian power sector poses a number of barriers. The utility industry is capital-intensive but has been suffering losses due to theft and subsidisation. The state-owned transmission and distribution companies have poor financial health and may not have the financial capacity to fund new technologies without the aid of the government.

No defined standards and guidelines exist for implementation of smart grid initiatives. The IEEE Standards Association (IEEE-SA) is working to partner with government and industry bodies in the country and engaging India's professional technical community, while leveraging its expertise and experience in global standards development to accelerate the process of establishing standards relevant to the Indian smart grid market.

Utilities and distribution companies have limited experience in communication technologies needed to manage critical components of the smart grid. Also, if India's utilities are to take an active part in designing and implementing IT and smart grid projects, they must acquire the skills required to take over operation and maintenance.

The most overlooked component of smart grid implementation is the education of consumers and what this means to them. It is important to build awareness to create an understanding of smart grids, the associated benefits and